

Local-Stochastic Volatility for Vanilla Modeling: A Tractable and Arbitrage Free Approach to Option Pricing

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First version: 4th May 2018
This version : 1st February 2019

Abstract

We present a novel and flexible technique for the construction of tractable, and fully arbitrage-free, Local-Stochastic Volatility (LSV) term distribution models. The method utilizes Lamperti's harmonic transform to combine a pure stochastic volatility (SV) process M_T with a general local volatility (LV) function σ . European options prices can conveniently be expressed in terms of a 1-dimensional integral over options on the SV component M_T . As a demonstration, we apply the technique to the popular SABR model family by letting M_T be the Normal SABR process, and letting the LV function σ be an extended CEV function, endowed with additional controls over the backbone and the wings for the smile. The resulting model not only rectifies the arbitrage problems of commonly used expansions for the SABR class, its additional parameters also allow for adequate handling of non-positive rates and improved calibration to quoted swaptions and CMS forward rates. Finally, for variety we also illustrate the technique using Heston dynamics and a general piecewise linear LV function.

1 Introduction

For the past two decades, LSV models have been enjoying a great deal of popularity amongst practitioners, not only as term structure models for the pricing and risk management of complex products (see for example Andersen & Andreasen[3], Blacher[9], Piterbarg[23] or Jaeckel and Kahl[19]), but also for the modeling of vanilla options (see for example Heston[16], Lipton[22], Beresticky, Busca and Florent[8] and the widely used SABR formula from Hagan et al[14]). Non-LSV models have also attracted some interest, e.g., the Levy process class (see Barndorff-Nielsen[7] and Ticot and

Charvet[26] for an application to Inflation modeling), the path-dependent LV model (see Guyon[12]) or direct parametrization of the implied volatility (see Gatheral[11]), but are used far less in practical applications.

In the world of vanilla interest rate derivatives (a particular area of focus for this article), standard LSV frameworks have recently been put under an increasing pressure from changing market environments, be it low or negative rates in major currencies or regulatory stress tests scenarios. To compound these difficulties, there are ever-growing demands on the models' ability to calibrate to a large number of swaptions while simultaneously capturing accurately the convexity embedded in CMS products. As a result, the near-standard SABR model volatility marking approach, based on the small-time asymptotics in Hagan et al[14], is now problematic, as the model is both insufficiently rich to match the market and also plagued by well-known arbitrage problems for high and low option strikes. A number of authors have proposed different approaches to tackle the current challenges. For example, Andreasen & Høge[5] use an extended SABR model based on a one step PDE method, producing an arbitrage-free interpolator, but with a term distribution that deviates from the original dynamics as time to maturity increases; Balland & Tran[6] propose a semi-analytical approach which is not guaranteed to be arbitrage-free (though it is stated that it will be in practical applications); and Antonov, Konikov & Spector[1][2] present a mixture approach.

While the improvements to the original SABR framework represent meaningful progress, none are, in our opinion, completely satisfactory from a theoretical or a practical perspective, and many involve either non-intuitive parametrization, possibility of arbitrage, lack of flexibility, or challenges in the implementation. Here, we present a new approach that allows to mix a generic LV function, parametric or non-parametric, with a pure SV process in an intuitive, tractable, easy-to-implement, and arbitrage-free way, in the sense that there are no negative butterflies for a given maturity. One design strength of the approach is that the choice of the LV component can be guided by criteria based on relevance rather than numerical and analytical feasibility.

The rest of the paper is organized as follows. In Section 2 we describe the LSV dynamics, present the fundamental idea of our method, and show how to handle the LV component (via a 1-dimensional integral), provided that the pure SV process is tractable, either via its Cumulative Density Function (CDF) or, equivalently, via its European options prices. In Section 3, we illustrate the method, pairing a general LV with a Normal SABR process. The Normal SABR dynamics are somewhat tractable, allowing for option prices through either a 2-dimensional integral of elementary functions (see Henry-Labordere[15], Ishihara[17] and Korn & Tang[20]) or as a 1-dimensional integral of special functions (see Antonov, Konikov & Spector[2]); unfortunately, the existing formulae become computationally too expensive when

embedded into our method and combined with a LV process. Thus, in Section 3.1 we develop new analytical formulae for option prices and probability densities in the Normal SABR model. These approximations are arbitrage-free by construction, computationally very simple (only a few calls to the Gaussian CDF are required), and accurate, even for maturities as long as 50 years. In Section 3.2 we make a concrete choice for the LV function and show in Section 3.3 that the model (which supports negative rates) can be successfully calibrated to a large strip of options and to CMS forward levels. In Section 4 we consider, for variety, an (arbitrary) piecewise linear LV combined with a Normal Heston. We describe the calibration strategy and provide option pricing formulae, as a discrete sum of options under the Heston model. Finally, Section 5 concludes the paper.

2 Local Stochastic Volatility

2.1 Notations and Dynamics

Our primary objective in this paper is to compute European option pricing on some forward process F_t , in the context of a general LSV model of the form

$$dF_t = \sigma(F_t)dM_t, \quad (1)$$

$$dM_t = \alpha_t dW_t. \quad (2)$$

In (1)-(2) the LV function σ , assumed positive, controls the support of the distribution and the dynamics of the model (also known as “backbone”). The pure SV component, M_t , is driven by a Brownian Motion W_t , with some generic volatility process α_t that will typically depend on a second source of randomness. In most model configurations, the skewness of implied volatilities as a function of strike are jointly determined by the form of σ and the correlation between W_t and α_t ; whereas the convexity of implied volatilities primarily originates with the variance of α_t and, in some case, with the convexity of σ . Popular choices for M_t included the log-normal dynamics of SABR, and the mean reverting CIR dynamics of the Heston process.

We emphasize that we are not focused on using (1)-(2) in a fully dynamic sense, as needed, for instance, for an arbitrage-free model for path-dependent interest rate options. Instead, we wish to use the model the same way that the SABR model is used in practice: as a model for marking European option prices at single maturities, likely with different model parameters for each maturity. This will ultimately affect how we treat certain drift terms that arise in theory, where our focus is on preserving only time 0 forwards, rather than imposing hard inter-temporal martingale conditions.

2.2 Local Volatility: a Reduced-Form approach

In this section, we show how to reduce the pricing of the LSV model into independent components via a replication argument, and present approximate options pricing formulae (4) and (5) in terms of 1-dimensional integrals. (either using Call prices C^M and Put prices P^M or using Φ^M).

2.2.1 Lamperti's Harmonic Transform and Drift Approximation

For a positive and sufficiently regular LV function σ , we consider the Lamperti transform

$$G(f) = \int_{F_0}^f \frac{du}{\sigma(u)},$$

and the associated process $g_t \triangleq G(F_t)$. An application of Ito's Lemma to g_t results in:

$$\begin{aligned} g_T &= M_T - \mu_T, \\ \mu_T &= \frac{1}{2} \int_0^T \alpha_t^2 \sigma'(G^{-1}(g_t)) dt. \end{aligned}$$

The stochastic drift μ_T ensures that $F_T = G^{-1}(g_T)$ is a martingale. As explained earlier, we are solely concerned with the term distribution of F_T for a given horizon T , so we make the pragmatic assumption that $\mu_T = \mu$, where μ is an ad-hoc constant, fixed at a value that ensures preservation of the forward, in the sense that

$$\mathbb{E}(F_T) = \mathbb{E}(G^{-1}(M_T - \mu)) = F_0,$$

where $\mathbb{E}$ is the expectation operator under the pricing measure $\mathbb{Q}$.

Accordingly, we from now on adopt the following definition for the term distribution of F_T :

$$F_T = G^{-1}(M_T - \mu). \quad (3)$$

The simplifying assumption made on the stochastic drift μ_T impacts only mildly the distribution of the underlying, and qualitatively the model behaves similarly to the true LSV SDE. Of course, F_T in (3) will converge to the exact solution of the SDE as time to maturity decreases, consistently with Beresticky, Busca and Florent[8]. Per construction, (3) will produce a valid (i.e., non-negative) density for F_T , and is not restrictive in terms of the resulting probability distribution, since the LV σ , being generic, can absorb any differences when calibrated.

2.3 Option Pricing

For any given random variable Y_T observed at T , we throughout adopt the following notations for a call value, a put value, and the CDF:

$$\begin{aligned} C^Y(K) &= \mathbb{E}(Y_T - K)^+, \\ P^Y(K) &= \mathbb{E}(K - Y_T)^+, \\ \Phi^Y(K) &= \mathbb{E}(1_{Y_T \leq K}). \end{aligned}$$

Now, since $F_T = G^{-1}(M_T - \mu)$, it follows that $C^F(K)$ and $P^F(K)$ can be written in terms of call/put options on M_T , via a replication argument (see Breeden-Litzenberger[10] and Rouah[24]):

$$\begin{aligned} C^F(K) &= \sigma(K)C^M(G(K) + \mu) + \int_K^\infty \sigma'(k)C^M(G(k) + \mu) dk, \\ P^F(K) &= \sigma(K)P^M(G(K) + \mu) - \int_{-\infty}^K \sigma'(k)P^M(G(k) + \mu) dk. \end{aligned} \quad (4)$$

We may also write these equations in terms of ϕ^M , the CDF of M_T :

$$\begin{aligned} C^F(K) &= \int_K^\infty \{1 - \phi^M(G(k) + \mu)\} dk, \\ P^F(K) &= \int_{-\infty}^K \phi^M(G(k) + \mu) dk. \end{aligned} \quad (5)$$

See Appendix A for a proof. The integrals involved in (4) and (5) can be efficiently implemented, provided that we have either fast closed-form solutions for options on M_T (see Section 3 for application to SABR style models) or specific forms for the LV function σ (see Section 4 for application to Heston model with piecewise linear LV).

3 SABR LV model

In this section, we use the Normal SABR process as the SV building block M_t in our model. As mentioned earlier, efficient closed-form expressions for options on M_t are critical to the practical implementation of (4) or (5), so we proceed to present efficient, arbitrage-free approximation formulae for options prices and the CDF for Normal SABR dynamics.

3.1 Normal SABR

We recall the Normal SABR model dynamics sets $F_t = M_t$, where

$$\begin{aligned} dM_t &= \alpha_t dW_t, \\ d\alpha_t &= \nu \alpha_t (\rho dW_t + \sqrt{1 - \rho^2} dW_t^\perp), \end{aligned}$$

where $\nu > 0$, $\rho \in (-1, 1)$, and W_t and $W_t^\perp$ are two independent Brownian motions. From Gyongy[13], we know that there exists a unique pure LV process $\tilde{M}_t$ with the same marginal distributions as M_t . In Lemma 3.1 we describe explicitly the dynamics of $\tilde{M}_t$.

Lemma 3.1. *Consider the process z_t , defined by*

$$dz_t = \nu \sqrt{1 + 2\rho z_t + z_t^2} dW_t, \quad z_0 = 0.$$

Then, for any $t \geq 0$, $\tilde{M}_t = M_0 + \frac{\alpha_0}{\nu} z_t$ has the same marginal distributions as M_t .

Proof. (Sketch) By the result of Gyongy, we know that the squared volatility term in the process $\tilde{M}_t$ is $\mathbb{E}(\alpha_t^2 | M_t = x)$, ensuring consistent marginal distributions between $\tilde{M}_t$ and M_t . An explicit calculation of this expression can be performed by a measure change, along the lines of Balland and Tran[6]. $\square$

Let us define $\xi_T \triangleq e^{\nu h_T}$, where

$$h_T = \int_0^{z_T} \frac{dz}{\nu \sqrt{1 + 2\rho z + z^2}}.$$

Lemma 3.2 below relates options on M_T to options on ξ_T .

Lemma 3.2. *For a given strike K , let $z^K \triangleq \frac{\nu}{\alpha_0}(K - M_0)$ and set*

$$\xi^K \triangleq \frac{\sqrt{1 - \rho^2 + (z^K + \rho)^2} + z^K + \rho}{1 + \rho}.$$

We then have

$$\begin{aligned} C^M(K) &= \frac{\alpha_0}{2\nu} \left[(1 + \rho) C^\xi(\xi^K) + (1 - \rho) P^\xi\left(\frac{1}{\xi^K}\right) \right], \\ \Phi^M(K) &= \Phi^\xi(\xi^K). \end{aligned} \quad (6)$$

Proof. Integrating explicitly the relation $h_T = \int_0^{z_T} \left(\nu \sqrt{1 + 2\rho z + z^2} \right)^{-1} dz$ and inverting the resulting expression leads to the reconstruction formula

$$z_T = -\rho + \frac{1}{2} \left((1 + \rho) \xi_T - \frac{1 - \rho}{\xi_T} \right). \quad (7)$$

From the definitions of z^K and ξ^K , we have:

$$z^K = -\rho + \frac{1}{2} \left((1 + \rho) \xi^K - \frac{1 - \rho}{\xi^K} \right).$$

Thus, subtracting z^K from z_T and reverting back to the initial variable $\tilde{M}_T$, we see that

$$M_T - K \stackrel{d}{=} \frac{\alpha_0}{2\nu} \left[(1 + \rho) (\xi_T - \xi^K) + (1 - \rho) \left(\frac{1}{\xi^K} - \frac{1}{\xi_T} \right) \right],$$

which shows that $M_T - K$ is distributed as a sum of two increasing functions in ξ_T vanishing simultaneously at ξ^K . Thus we clearly have

$$(M_T - K)^+ \stackrel{d}{=} \frac{\alpha_0}{2\nu} \left[(1 + \rho) (\xi_T - \xi^K)^+ + (1 - \rho) \left(\frac{1}{\xi^K} - \frac{1}{\xi_T} \right)^+ \right], \quad (8)$$

$$1_{M_t \leq K} \stackrel{d}{=} 1_{\xi_t \leq \xi^K}. \quad (9)$$

Applying the expectation operator $\mathbb{E}$ then proves the lemma. $\square$

We remark that the elegant technique used to prove (8)-(9) is known as *Jamshidian's Trick*, originally used in the context of swaption valuation in a one-factor model, see Jamshidian[18]. Our next goal is to work out accurate and non-arbitrageable closed-form approximations for C^ξ , P^ξ and Φ^ξ . We start by using Ito's Lemma to note that

$$dh_t = dW_t - \frac{\nu}{2} \tanh(\nu(h_t + \bar{h})) dt, \quad (10)$$

with

$$\bar{h} \triangleq \frac{1}{2\nu} \ln \left(\frac{1 + \rho}{1 - \rho} \right).$$

The dynamics of h_t in (10) resemble a mean-reverting drifted Brownian motion, so both its mean and variance will be affected by the (locally) stochastic drift term. Given that ξ_T is defined as an exponential of h_t , one is tempted to use a simple log-normal approximation for ξ_T , with moment-matched mean and variance. Indeed, we find that this works quite well, but here present an alternative in Proposition 3.3 with even better precision. The idea behind the proposition is to apply a measure change to capture more accurately the subtle dynamics of h_T , without sacrificing tractability.

Proposition 3.3. *The following (arbitrage-free) expressions can be used as very accurate closed-form solutions for option prices and the CDF for the process ξ_T :*

$$\begin{aligned} C^\xi(\xi^K) &\approx \frac{3\Gamma}{2\gamma} C(\nu, \nu, k, \bar{h}, T) - \frac{\Gamma}{2\gamma} C(\nu, 3\nu, k, \bar{h}, T), \\ P^\xi\left(\frac{1}{\xi^K}\right) &\approx \frac{1}{2\gamma\Gamma} C(-\nu, 3\nu, k, \bar{h}, T) - \frac{3}{2\gamma\Gamma} C(-\nu, \nu, k, \bar{h}, T), \\ 1 - \Phi^\xi(\xi^K) &\approx \frac{3}{2\gamma} \Upsilon(\nu, k, \bar{h}, T) - \frac{1}{2\gamma} \Upsilon(3\nu, k, \bar{h}, T), \end{aligned}$$

with

$$k = \frac{1}{\nu} \ln \left(\frac{\xi^K}{\Gamma} \right), \quad (11)$$

$$\gamma = \frac{3}{2} \Upsilon^*(\nu, \bar{h}, T) - \frac{1}{2} \Upsilon^*(3\nu, \bar{h}, T), \quad (12)$$

$$\Gamma = \frac{\rho + \sqrt{\rho^2 + (1 - \rho^2)E^+E^-}}{(1 + \rho)E^+}, \quad (13)$$

$$E^\pm = \frac{e^{\frac{\nu^2 T}{2}}}{2\gamma} (3\Upsilon^*(\nu, \bar{h} \pm \nu T, T) - \Upsilon^*(3\nu, \bar{h} \pm \nu T, T)), \quad (14)$$

and where the auxiliary functions Υ^* , Υ and C have been defined below via the Gaussian kernel and re-expressed, using elementary integral manipulations, in terms of the Normal CDF, $N(\cdot)$:

$$\begin{aligned} \Upsilon^*(\lambda, \bar{h}, T) &\triangleq \int_{-\infty}^{+\infty} \frac{e^{-\frac{\lambda}{2}|h+\bar{h}|}}{\sqrt{2\pi T}} e^{-\frac{h^2}{2T}} dh \\ &= e^{\frac{\lambda^2 T}{8}} \left[e^{-\frac{\lambda \bar{h}}{2}} N\left(\frac{\bar{h}}{\sqrt{T}} - \frac{\lambda\sqrt{T}}{2}\right) + e^{\frac{\lambda \bar{h}}{2}} N\left(-\frac{\bar{h}}{\sqrt{T}} - \frac{\lambda\sqrt{T}}{2}\right) \right], \end{aligned} \quad (15)$$

$$\begin{aligned} C(\sigma, \lambda, k, \bar{h}, T) &\triangleq \int_k^{+\infty} (e^{\sigma h} - e^{\sigma k}) + \frac{e^{-\frac{\lambda}{2}|h+\bar{h}|}}{\sqrt{2\pi T}} e^{-\frac{h^2}{2T}} dh \\ &= e^{\frac{\sigma^2 T}{2}} \Upsilon(\lambda, k - \sigma T, \bar{h} + \sigma T, T) - e^{\sigma k} \Upsilon(\lambda, k, \bar{h}, T), \end{aligned} \quad (16)$$

with

$$\Upsilon(\lambda, k, \bar{h}, T) \triangleq \int_k^{+\infty} \frac{e^{-\frac{\lambda}{2}|h+\bar{h}|}}{\sqrt{2\pi T}} e^{-\frac{h^2}{2T}} dh, \quad (17)$$

$$= e^{\frac{\lambda^2 T}{8} - \frac{\lambda \bar{h}}{2}} N\left(-\frac{k}{\sqrt{T}} - \frac{\lambda\sqrt{T}}{2}\right) \text{ if } k \geq -\bar{h}, \quad (18)$$

$$= \Upsilon^*(\lambda, \bar{h}, T) - e^{\frac{\lambda^2 T}{8} + \frac{\lambda \bar{h}}{2}} N\left(\frac{k}{\sqrt{T}} - \frac{\lambda\sqrt{T}}{2}\right) \text{ if } k < -\bar{h}. \quad (19)$$

Proof. The proof relies in part on results listed in Appendix B. We consider the exponential martingale θ_t^* in the measure $\mathbb{Q}$:

$$\frac{d\theta_t^*}{\theta_t^*} = \frac{\nu}{2} \tanh(\nu(h_t + \bar{h})) dW_t,$$

and the associated measure $\mathbb{Q}^*$ defined via $\frac{d\mathbb{Q}^*}{d\mathbb{Q}} = \theta_T^*$. By construction, h_T is a driftless Brownian Motion under $\mathbb{Q}^*$. Using loose notation,

$$\mathbb{E}(1_{h_T=h}) = \mathbb{E}^*\left(\frac{1}{\theta_T^*} 1_{h_T=h}\right) = \mathbb{E}^*\left(\mathbb{E}^*\left[\frac{1}{\theta_T^*} | h_T\right] 1_{h_T=h}\right),$$

where $\mathbb{E}^*$ denotes expectation in measure $\mathbb{Q}^*$. Defining $f(h) = \sqrt{\cosh(\nu(h + \bar{h}))}$ and applying Ito's Lemma, we find that $d(f(h_t)/\theta_t^*) = O(dt)$. Moreover, using a Taylor expansion to second order in $e^{-\nu|h+\bar{h}|}$, we see that

$$\frac{1}{f(h)} \approx e^{-\frac{\nu}{2}|h+\bar{h}|} \frac{3 - e^{-\nu|h+\bar{h}|}}{2}.$$

Thus, we can write

$$\begin{aligned} \mathbb{E}^* \left[\frac{1}{\theta_T^*} | h_T \right] &= \frac{1}{f(h_T)} \mathbb{E}^* \left[\frac{f(h_T)}{\theta_T^*} | h_T \right], \\ &\approx \frac{1}{\gamma} e^{-\frac{\nu}{2}|h_T+\bar{h}|} \frac{3 - e^{-\nu|h_T+\bar{h}|}}{2} \triangleq \frac{1}{\theta^\dagger(h_T)}, \end{aligned}$$

The constant γ is introduced to ensure that $E^{Q^*} \left(\frac{1}{\theta^\dagger(h_T)} \right) = 1$. Finally, we can write $\xi_T = e^{\nu h_T} \approx \Gamma e^{\nu h_T^\dagger}$ where the density of $h_T^\dagger$ under Q is:

$$\varrho(h) = \frac{e^{-\frac{\nu}{2}|h+\bar{h}|} \left(3 - e^{-\nu|h+\bar{h}|} \right)}{2\gamma} \frac{e^{-\frac{h^2}{2T}}}{\sqrt{2\pi T}}. \quad (20)$$

γ can be found by enforcing $\int_{-\infty}^{+\infty} \varrho(h) dh = 1$, see (B.1). The parameter Γ , which has been introduced to account for the small measure change approximations made on the way, can be found from the martingale condition $E(z_T) = 0$ in (7), see (B.2). Using results of Appendix B leads to the expressions listed in the Lemma for γ , Γ and $E^\pm$, via the function Υ^* , defined at (15). Note that (20) is a valid density by construction and completely tractable. As a matter of fact, as shown in the Proposition, C^ξ , $P_\xi^\frac{1}{2}$ and Φ^ξ can be computed explicitly with only few Calls to the Gaussian CDF; see also (B.3) for details. $\square$

Bringing together results from Lemma 3.2 and Proposition 3.3, we arrive at explicit and fast approximations for options on M_T . We emphasize that these expressions are arbitrage-free by construction, for any set of parameters and any maturity. Empirically, the approximations perform surprisingly well, even for large maturities, as testified by the results shown in Figure 1 which compares our approximation to both the Hagan expansion in Hagan et al[14] and to exact values in Antonov et al[2] (computed by an expensive integral-based closed form solution). The new formulae only require a few calls to the Gaussian CDF, and has similar computational performance as the Hagan expansion.

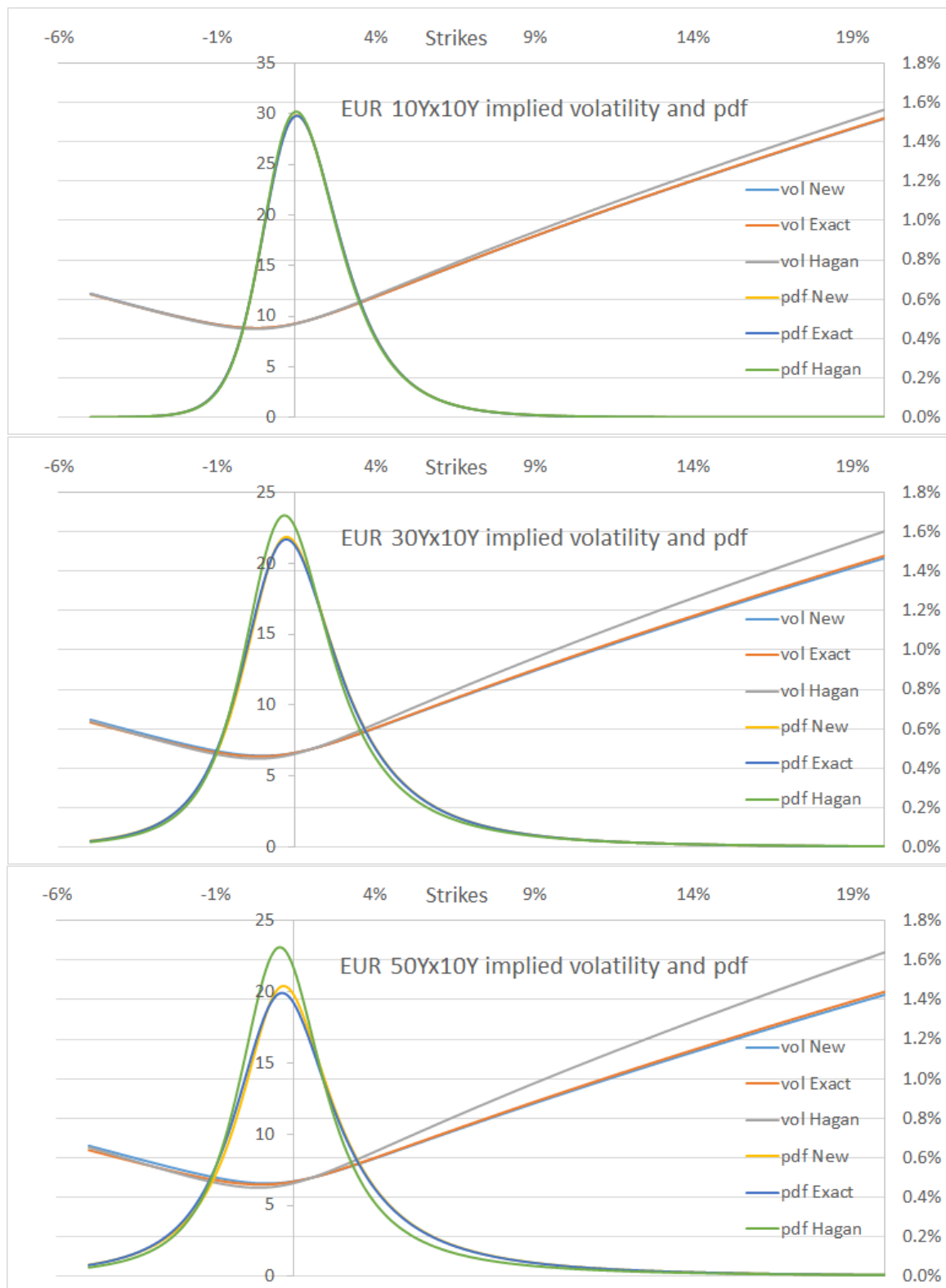


Figure 1: Comparison of implied PDF (left vertical axis) and normal implied volatility (right vertical axis) for our new approach (“New”), the exact approach in Antonov et al[2] (“Exact”), and the asymptotic approximation in Hagan et al[14] (“Hagan”) for the Normal SABR model. The swaption tenor is 10 years, and maturities vary between 10 and 50 years, as shown in the figures. Forward $F_0 = 2\%$, ATM normal Vol $\hat{\sigma} = 0.5\%$, Vol-of-vol $\nu = 20\%$ and correlation $\rho = 50\%$.

3.2 Local Volatility Specification

Now that the problem of working out fast and accurate approximations for the Normal SABR model is out of the way, we can freely design the LV function σ in (1) to satisfy a set of desired features, for instance:

- Adequate support; e.g. distribution for F_T should allow for negative rates.
- Flexibility; the parametrization needs to be generic enough to be calibrated to a reasonably large number of liquid option strikes (e.g., 5).
- Control over dynamics (back-bone); should comply with trader requirements and empirical evidence.
- Control over model behavior at high and low strikes and, ultimately, of convex product like CMS.

While many different LV specifications could be used to target these requirements, let us give reasonable example for practical purposes. For instance, to address the first point, we choose an LV function that remains positive for negative values of F . To this aim, for $\epsilon \geq 0$, we introduce the function

$$\eta(\epsilon, f) = \epsilon \ln(1 + e^{\frac{f}{\epsilon}})$$

that regularizes f^+ . To control the backbone, we consider a CEV (power) specification, with level-dependent power:

$$\beta(f) = \frac{1}{2}(\beta_l + \beta_h) + \frac{1}{2}(\beta_h - \beta_l) \tanh(\lambda(f - F_0)).$$

Note that $\beta(f)$ varies smoothly from β_l to β_h , with a speed controlled by the constant λ . So-equipped, we are now able to explicitly define the LV function:

$$\sigma^*(f) = (\eta(\epsilon, f))^{\beta(f)} + \psi \eta(\epsilon_s, f - K_h)$$

In the function σ^* , the parameter ϵ controls the smile for low strikes and typically the probability mass below 0. If $\epsilon = 0$, then F_T remains above 0 (if started above 0). K_h is an arbitrary strike outside the region of observable strikes, but beyond which the smile still impacts CMS forwards. The parameter ψ controls the volatility beyond K_h and the parameter ϵ_s is used as a smoothing parameter. To avoid explosive behavior of σ^* in the wings, we finally choose to extrapolate the LV flat outside some interval $[\underline{f}, \bar{f}]$. One benefit of this extrapolation is that the integrals in (4) are truncated over finite support, making them more amenable to fast numerical integration. For interest rate application, a possible parameterization for $\underline{f}$, $\bar{f}$ and K_h is shown in Figure 2. The final expression for the LV reads:

$$\sigma(f) = \sigma^*(\min(\max(f, \underline{f}), \bar{f})) \quad (21)$$

Below we demonstrate the impact of the parameters controlling the wings of the LV.

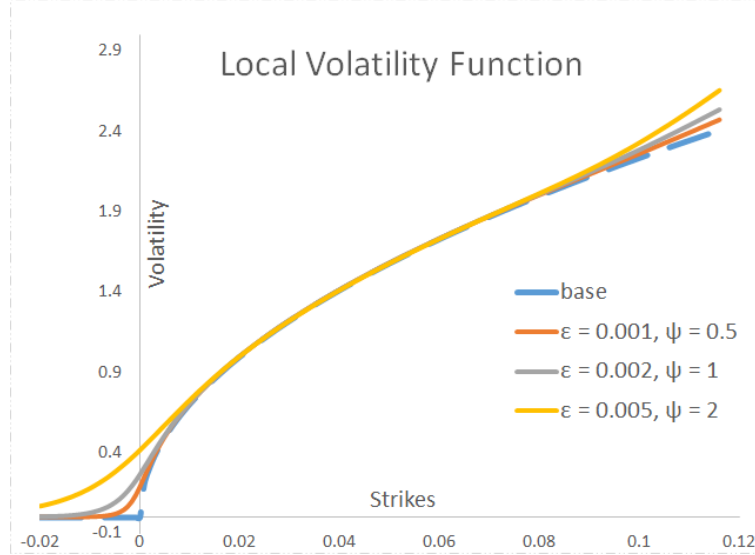


Figure 2: Control over the wings of the LV (normalized at the forward $F_0 = 0.02$) varying ϵ and Ψ . Base case (square-root LV) corresponds to $\beta_l = \beta_h = \frac{1}{2}$, $\epsilon = 0$, $\epsilon_s = 0.01$, $\underline{f} = F_0 - 3\%$, $\bar{f} = F_0 + 50\%$, $K_h = F_0 + 8\%$ and $\Psi = 0$.

3.3 Model Settings and Numerical results

We consider the LV function σ defined in (21) via a set of LV parameters and the associated Lamperti transform $G(f) = \int_{F_0}^f \sigma(u)^{-1} du$. $G(f)$ can be computed and cached, once and for all, using for instance a trapezoid quadrature rule on a fine grid. Options on F_T struck at K are then expressed in (4) where C^M is computed using Lemma 3.2 and Proposition 3.3; put options P^M may be computed using put-call parity.

For some fixed maturity T , our first task is calibrating μ and α_0 to simultaneously reprice the forward (or equivalently ensuring $C^F(F_0) = P^F(F_0)$) and the at-the-money (ATM) option, $C^F(F_0) = \hat{\sigma} \sqrt{T/(2\pi)}$, where $\hat{\sigma}$ is the Normal (Bachelier) ATM implied volatility. This is achieved using the following expressions for ATM call and put values, derived from (6):

$$C^F(F_0) = \sigma(F_0)C^M(\mu) + \int_{F_0}^{\infty} \sigma'(k)C^M(G(k) + \mu) dk,$$

$$P^F(F_0) = \sigma(F_0)P^M(\mu) - \int_{-\infty}^{F_0} \sigma'(k)P^M(G(k) + \mu) dk.$$

Note that, once calibrated, the method allows to price simultaneously a strip of options by adequately discretizing the integrals in (4). In Section

3.3.1, the model is degenerated to a regularized SABR square-root model and compared to the Hagan asymptotic formula. In Section 3.3.2 the model is (successfully) calibrated to market quotes (strip of options and CMS forward).

3.3.1 SABR Square root

In this section we consider, as an example, the SABR square root dynamics by setting:

$$\begin{aligned} dF_t &= \alpha_t \sqrt{F_t} dW_t, \\ d\alpha_t &= \nu \alpha_t (\rho dW_t + \sqrt{1 - \rho^2} dW_t^\perp). \end{aligned}$$

In Figure 3 we compare our new approach (using the LV defined in Equation 21) with a CEV power of $\frac{1}{2}$ and a smoothing parameter $\epsilon = 0.001$ to remove the singularity at the origin) to the standard Hagan expansion applied to the square-root SABR model, where both models are calibrated to the ATM normal vol $\hat{\sigma}$. We see that the volatility skew in the new model is close to the one of the Hagan formula for strikes around the forward, but is much smoother, and with no negative densities, at low strikes.

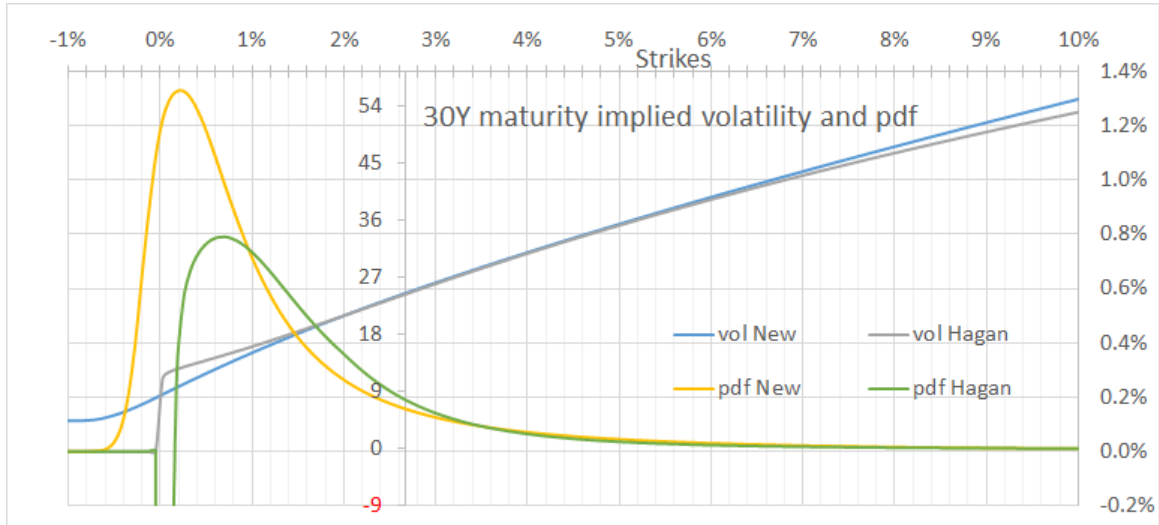


Figure 3: Comparison of a 30Y maturity implied PDF (left vertical axis) and implied Normal volatility (right vertical axis) between our new approach (“New”) and Hagan et al [14] approximation (“Hagan”) for the square root SABR model. Forward $F_0 = 2\%$, $\hat{\sigma} = 0.5\%$, Vol-of-vol $\nu = 20\%$, correlation $\rho = 50\%$ and $\beta_l = \beta_h = \frac{1}{2}$. Smoothing near 0 is achieved using $\epsilon = 0.001$.

3.3.2 Calibration to Market Instruments

In this section, our model is used with a full set of LV and SV parameters, calibrated to best-fit option prices and CMS forwards seen in market quotes. In Figure 4, we display the PDF and the Normal (Bachelier) volatility implied by the model, using the Normal SABR formula (6) and the full LV function (21). We can see that a very accurate calibration to these instruments can be achieved, with the resulting implied PDF being smooth and well-behaved.

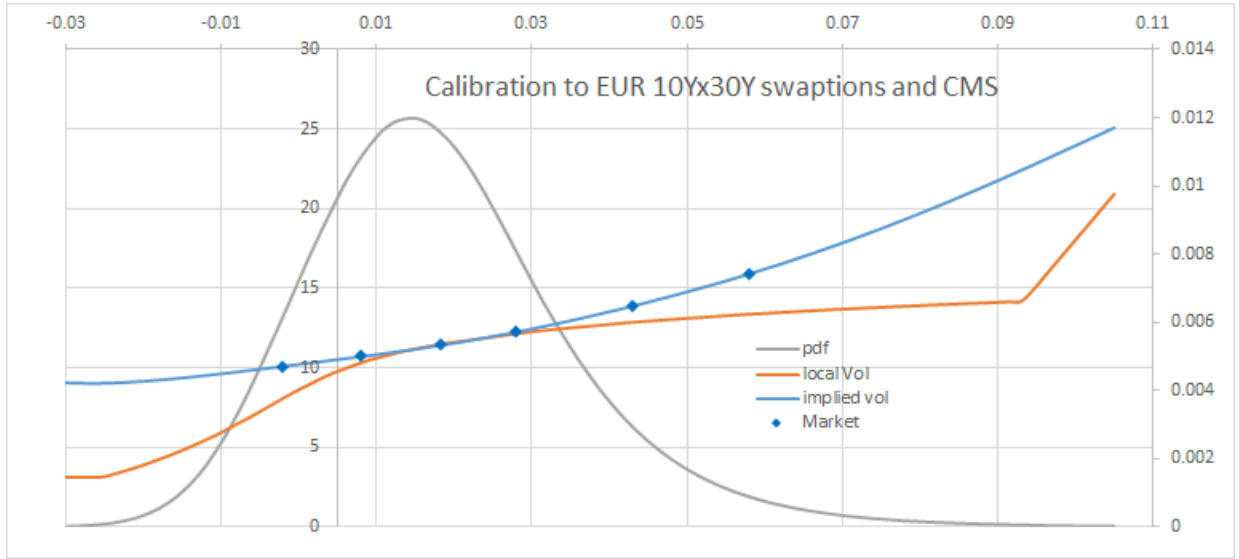


Figure 4: implied PDF (left vertical axis), LV and implied Normal volatility (right vertical axis) for a model calibrated to swaptions implied volatilities (Market) and the CMS convexity adjustments (61 basis points). Forward $F_0 = 1.833\%$, ATM Vol $\hat{\sigma} = 0.533\%$, Vol of vol $\nu = 16.5\%$, correlation $\rho = 7\%$, $\beta_l = 26.5\%$, $\beta_h = 0\%$, $\epsilon = 0.0032$, $\epsilon_s = 0.001$, $\lambda = 0.5$ and $\psi = 47.7$

4 Heston Model with Linear Piecewise Local Volatility

As mentioned earlier, the proposed method for building LSV term volatility models is generic and can accommodate different dynamics. In this section, we briefly illustrate the breath of the model, by using it to equip the Heston Model with a general piecewise linear LV. For this purpose, we consider, as a building block, a Normal Heston dynamics for the process M_t :

$$\begin{aligned} dM_t &= \sqrt{\zeta_t} dW_t, \\ d\zeta_t &= \kappa(\theta - \zeta_t)dt + \nu\sqrt{\zeta_t}(\rho dW_t + \sqrt{1 - \rho^2} dW^\perp), \end{aligned}$$

where ζ_0 , ρ and ν control respectively the ATM level, the skew, and the convexity of the implied volatility of the model.

The characteristic function of M_T , $\chi(u) = E(e^{iuM_T})$ is known in closed form (see Heston[16]), thus the Lewis-Lipton formula can be employed to recover option prices on M_T or on $e^{\lambda M_T}$, via an inverse Fourier integral (see Lipton[22], Lewis[21] or Andersen & Piterbarg[4]). The general pricing equation (4) would not be computationally workable, as it would involve double integrals. Nevertheless, in the next section we show how the assumption of piecewise linearity on the LV allows to write options prices as a discrete sum of options in the standard Heston model.

For later use, let us denote

$$V_{c,p}^H(s, G) = \mathbb{E} \left(\epsilon_{c,p} \frac{e^{s(M_T - \mu - G)} - 1}{s} \right)^+,$$

with the convention $V_{c,p}^H(0, G) = E(\epsilon_{c,p}(M_T - \mu - G))^+$, $\epsilon_c = 1$ (resp. $\epsilon_p = -1$) for a call (resp. for a put).

4.1 Replication Formula using Calls and Puts

Assuming now a piecewise linear LV (extrapolated flat, for concreteness), we consider a strip of n strikes $K_{1 \leq i \leq n}$; without loss of generality, we can assume that the forward $F_0 = K_p$, with $p \in]1, n[$. We also define the associated volatilities $\sigma_{1 \leq i \leq n}$. Let $\sigma'_i = \frac{\sigma_{i+1} - \sigma_i}{K_{i+1} - K_i}$ be the LV slopes, and then define the LV function

$$\sigma(f) = \sigma_1 1_{f < K_1} + \sum_{i=1}^{n-1} (\sigma_i + \sigma'_i(f - K_i)) 1_{K_i \leq f < K_{i+1}} + \sigma_n 1_{K_n \leq f}.$$

We also denote $G_i = G(K_i)$ as the (Lamperti) transformed variables on the strike grid. The following identities hold:

$$\begin{aligned} P^F(K_1) &= \sigma_1 V_p^H(0, G_1), \\ P^F(K_{i+1}) - P^F(K_i) &= \sigma_{i+1} V_p^H(\sigma'_i, G_{i+1}) - \sigma_i V_p^H(\sigma'_i, G_i), \quad 1 \leq i < p, \\ C^F(K_i) - C^F(K_{i+1}) &= \sigma_i V_c^H(\sigma'_i, G_i) - \sigma_{i+1} V_c^H(\sigma'_i, G_{i+1}), \\ C^F(K_n) &= \sigma_n V_c^H(0, G_n), \quad p \leq i < n; \end{aligned} \tag{22}$$

see (C) for a proof.

4.2 Calibration

The calibration problem involves the constraints of re-pricing the forward and reproducing the ATM volatility $\hat{\sigma}$. The two parameters available for

this task are the drift μ and the initial value of the volatility, ζ_0 .

$$P^F(F_0) = \sigma_1 V_p^H(0, G_1) + \sum_{i=1}^{p-1} \sigma_{i+1} V_p^H(\sigma'_i, G_{i+1}) - \sigma_i V_p^H(\sigma'_i, G_i),$$

$$C^F(F_0) = \sigma_n V_c^H(0, G_n) + \sum_{i=p}^{n-1} \sigma_i V_c^H(\sigma'_i, G_i) - \sigma_{i+1} V_c^H(\sigma'_i, G_{i+1}).$$

Thus the calibration problem can be written as the system

$$P^F(F_0) - C^F(F_0) = 0,$$

$$P^F(F_0) + C^F(F_0) = \hat{\sigma} \sqrt{\frac{2T}{\pi}},$$

which can be solved in μ and ζ_0 by any standard 2-dimensional root search algorithm.

4.3 Option Pricing

Consider a Call struck at $K \in [K_i, K_{i+1}]$. We have:

$$C^F(K) = C^F(K_{i+1}) + \sigma(K) V_c^H(\sigma'_i, G(K)) - \sigma_{i+1} V_c^H(\sigma'_i, G_{i+1}).$$

Note that $C^F(K_{i+1})$ has already been computed at the calibration stage. Thus, it is straightforward to price an option on F_T struck at K . Note that, in the degenerate case where the Normal Heston process M_t is Gaussian ($\nu = 0$), the model can be used as an arbitrage-free LV interpolator, in the same line as what has been done in Schlenkrich[25].

5 Conclusion

We have presented a new method for term distribution modeling that allows the user mix an arbitrary LV with an arbitrary SV in a tractable, intuitive and arbitrage-free way. We have applied the method to the SABR models family, pairing a flexible LV with a Normal SABR SV model. In the process, we developed new efficient closed form option pricing formulae for the Normal SABR SV model. The resulting LSV model is smooth and arbitrage-free, and successfully calibrates to market quotes for swaptions and CMS forwards. Finally, we have also shown how the method can be used in the context of the Heston model, by providing closed-form pricing solutions when the LV is a general linear piecewise function and the SV is a Normal Heston model.

Acknowledgement

I am indebted to Leif Andersen and Guillaume Blacher for insightful suggestions, comments and support. I am very grateful to Mark Lake for an efficient implementation of the model, and want to thank Guillaume Royer, Julien Pantz and Elias Daboussi for fruitful discussions. Finally thanks to Ming Zhu and Mike Staunton for spotting few typos.

A Replication

For any twice-differentiable function f , we have the identity (due to Carr-Madan and available in Rouah[24]):

$$f(S) - f(a) = (S - a)f'(a) + \int_{-\infty}^a (u - S)^+ f''(u) du + \int_a^{+\infty} (S - u)^+ f''(u) du.$$

Using $f = G^{-1}$, $S = M_T - \mu$, $a = G(K)$ and observing that $(G^{-1})' \circ G = \sigma$ and $(G^{-1})'' \circ G = \sigma' \sigma$, we have:

$$\begin{aligned} F_T - K &= \sigma(K) (M_T - \mu - G(K)) + \int_K^{+\infty} (M_T - \mu - G(k))^+ \sigma'(k) dk \\ &\quad + \int_{-\infty}^K (G(k) + \mu - M_T)^+ \sigma'(k) dk. \end{aligned}$$

Multiplying the last equality by $1_{K \leq F_T}$, we obtain:

$$\begin{aligned} (F_T - K)^+ &= \sigma(K) (M_T - \mu - G(K))^+ \\ &\quad + \int_K^{+\infty} \sigma'(k) (M_T - \mu - G(k))^+ dk. \end{aligned}$$

Finally, applying the expectation operator $\mathbb{E}$ to the last identity leads to (4); the formulae for puts is similar.

To prove (5) we simply notice that, for puts:

$$(K - F_T)^+ = \int_{-\infty}^K 1_{F_T \leq k} dk = \int_{-\infty}^K 1_{G(F_T) \leq G(k)} dk = \int_{-\infty}^K 1_{M_T \leq G(k) + \mu} dk,$$

and similarly for calls. Applying the expectation operator yields the desired result.

B Normal SABR

B.1 Normalizing Constant γ

$$\begin{aligned}\gamma &\triangleq \int_{-\infty}^{+\infty} e^{-\frac{\nu}{2}|h+\bar{h}|} \left(\frac{3 - e^{-\nu|h+\bar{h}|}}{2} \right) \frac{1}{\sqrt{2\pi T}} e^{-\frac{h^2}{2T}} dh \\ &= \frac{3}{2} \int_{-\infty}^{+\infty} e^{-\frac{\nu}{2}|h+\bar{h}|} \frac{e^{-\frac{h^2}{2T}}}{\sqrt{2\pi T}} dh - \frac{1}{2} \int_{-\infty}^{+\infty} e^{-\frac{3\nu}{2}|h+\bar{h}|} \frac{e^{-\frac{h^2}{2T}}}{\sqrt{2\pi T}} dh \\ &= \frac{3}{2} \Upsilon^*(\nu, \bar{h}, T) - \frac{1}{2} \Upsilon^*(3\nu, \bar{h}, T),\end{aligned}$$

where Υ^* has been defined in (15).

B.2 Computation of Γ

In order to reprice the forward, we must have $E(z_T) = 0$ in (7). That is:

$$\begin{aligned}0 &= -\rho + \frac{1}{2} \left((1 + \rho) \Gamma \mathbb{E}(e^{\nu h_T}) - \frac{1 - \rho}{\Gamma} \mathbb{E}(e^{-\nu h_T}) \right) \\ &= -\rho + \frac{1}{2} \left((1 + \rho) \Gamma E^+ - \frac{1 - \rho}{\Gamma} E^- \right),\end{aligned}$$

where

$$E^\pm \triangleq \int_{-\infty}^{+\infty} e^{\pm \nu h} \varrho(h) dh.$$

This leads to a second-order equation in Γ , with admissible solution:

$$\Gamma = \frac{\rho + \sqrt{\rho^2 + (1 - \rho^2) E^+ E^-}}{(1 + \rho) E^+}.$$

We have:

$$\begin{aligned}E^\pm &= \int_{-\infty}^{+\infty} e^{\pm \nu h} \frac{e^{-\frac{\nu}{2}|h+\bar{h}|} \left(\frac{3 - 2e^{-\nu|h+\bar{h}|}}{2} \right)}{\gamma} \frac{e^{-\frac{h^2}{2T}}}{\sqrt{2\pi T}} dh \\ &= e^{\frac{\nu^2 T}{2}} \int_{-\infty}^{+\infty} \frac{e^{-\frac{\nu}{2}|h+\bar{h}|} \left(\frac{3 - 2e^{-\nu|h+\bar{h}|}}{2} \right)}{\gamma} \frac{e^{-\frac{(h \mp \nu T)^2}{2T}}}{\sqrt{2\pi T}} dh \\ &= e^{\frac{\nu^2 T}{2}} \int_{-\infty}^{+\infty} \frac{e^{-\frac{\nu}{2}|h+\bar{h} \pm \nu T|} \left(\frac{3 - 2e^{-\nu|h+\bar{h} \pm \nu T|}}{2} \right)}{\gamma} \frac{e^{-\frac{h^2}{2T}}}{\sqrt{2\pi T}} dh \\ &= \frac{e^{\frac{\nu^2 T}{2}}}{2\gamma} (3\Upsilon^*(\nu, \bar{h} \pm \nu T, T) - \Upsilon^*(3\nu, \bar{h} \pm \nu T, T)),\end{aligned}$$

with Υ^* defined in (15).

B.3 Options Prices on ξ and $\frac{1}{\xi}$

Since $\xi_T = \Gamma e^{\nu h_T}$, we have:

$$\begin{aligned} C^\xi(\xi^K) &= \mathbb{E}(\xi_T - \xi^K)^+ = \Gamma \int_k^{+\infty} (e^{\nu h} - e^{\nu k}) \varrho(h) dh \\ &= \frac{3\Gamma}{2\gamma} C(\nu, \nu, k, \bar{h}, T) - \frac{\Gamma}{2\gamma} C(\nu, 3\nu, k, \bar{h}, T), \\ P^{\frac{1}{\xi}}\left(\frac{1}{\xi^K}\right) &= \mathbb{E}\left(\frac{1}{\xi^K} - \frac{1}{\xi_T}\right)^+ = \frac{1}{\Gamma} \int_k^{+\infty} (e^{-\nu k} - e^{-\nu h}) \varrho(h) dh \\ &= \frac{1}{2\gamma\Gamma} C(-\nu, 3\nu, k, \bar{h}, T) - \frac{3}{2\gamma\Gamma} C(-\nu, \nu, k, \bar{h}, T), \end{aligned}$$

where k has been defined in (11) and the function C has been defined in (16).

C Heston Piecewise Linear

On each sub-intervals $]K_i, K_{i+1}]$, the LV is piecewise linear, and extrapolated flat in $] -\infty, K_1]$ and $]K_N, +\infty]$.

$$\begin{aligned} (G(K_1) - G(F_T))^+ &= 1_{F_T \leq K_1} \int_{F_T}^{K_1} \frac{dk}{\sigma(k)} \\ &= \frac{(K_1 - F_T)^+}{\sigma_1}, \end{aligned}$$

thus

$$(K_1 - F_T)^+ = \sigma_1 (G(K_1) - G(F_T))^+ = \sigma_1 (G_1 + \mu - M_T)^+.$$

For $1 \leq i < p$

$$(K_{i+1} - F_T)^+ - (K_i - F_T)^+ = (K_{i+1} - G^{-1}(g_T))^+ - (K_i - G^{-1}(g_T))^+,$$

which depends only on the value of $G^{-1}(g_T)$ for $G_i \leq g_T \leq G_{i+1}$. Thus, we can assume $K_i \leq f \leq K_{i+1}$ for the purpose of computing G :

$$G(f) = G_i + \int_{K_i}^f \frac{dk}{\sigma_i + \sigma'_i(k - K_i)} = G_i + \frac{1}{\sigma'_i} \ln \left(1 + \frac{\sigma'_i}{\sigma_i} (f - K_i) \right).$$

Thus:

$$\begin{aligned} K_i - G^{-1}(g) &= \sigma_i \frac{1 - e^{\sigma'_i(g - G_i)}}{\sigma'_i}, \\ K_{i+1} - G^{-1}(g) &= \sigma_{i+1} \frac{1 - e^{\sigma'_i(g - G_{i+1})}}{\sigma'_i}. \end{aligned}$$

Finally we have, for $1 \leq i < p$,

$$(K_1 - F_T)^+ = \sigma_1 (G_1 + \mu - M_T)^+ \\ (K_{i+1} - F_T)^+ - (K_i - F_T)^+ = \sigma_{i+1} \left(\frac{1 - e^{\sigma'_i(M_T - \mu - G_{i+1})}}{\sigma'_i} \right)^+ - \sigma_i \left(\frac{1 - e^{\sigma'_i(M_T - \mu - G_i)}}{\sigma'_i} \right)^+.$$

For $p \leq i < n$, the proof is similar:

$$(F_T - K_i)^+ - (F_T - K_{i+1})^+ = \sigma_i \left(\frac{e^{\sigma'_i(M_T - \mu - G_i)} - 1}{\sigma'_i} \right)^+ - \sigma_{i+1} \left(\frac{e^{\sigma'_i(M_T - \mu - G_{i+1})} - 1}{\sigma'_i} \right)^+, \\ (F_T - K_n)^+ = \sigma_n (M_T - \mu - G_n)^+.$$

Applying the expectation operator $\mathbb{E}$, we get (22).

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